

B.Tech. Degree VI Semester Examination April 2018

CS 15-1603 CRYPTOGRAPHY AND NETWORK SECURITY (2015 Scheme)

Time: 3 Hours

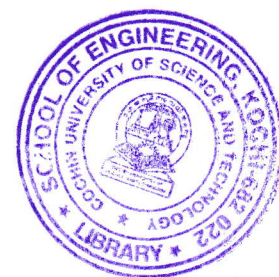
Maximum Marks: 60

PART A

(Answer **ALL** questions)

(10 × 2 = 20)

- I. (a) What are the key objectives of computer security?
- (b) With the help of play fair cipher encrypt the message *Have His Carcase* using the key *monarchy*.
- (c) What are the dimensions with which cryptographic systems are classified?
- (d) What are the parameters with which Feistel cipher depends?
- (e) What is the need of S box in AES encryption?
- (f) Discuss the role of Fermat's theorem and Euler's theorem in the development of RSA algorithm.
- (g) Check whether the number 221 is prime using primality testing algorithm.
- (h) Illustrate Diffie-Hellman key exchange using elliptic curve arithmetic.
- (i) What is the generic model of digital signature?
- (j) What are the features and requirements of Kerberos?



PART B

(4 × 10 = 40)

- II. (a) Explain the OSI security architecture. (5)
 - (b) Use Hill cipher to encrypt the message 'paymoremoney' with the key (5)
- $$K = \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$$

OR

- III. Define group, ring, field with examples. Also define the finite field GF(p) and GF(pⁿ). (10)
- IV. Explain DES algorithm and its strength. (10)
- OR
- V. Discuss the structure of AES encryption and decryption process. (10)
- VI. Discuss ElGamal encryption and decryption. Encrypt the message M = 17 with q = 19. (10)
- OR
- VII. Explain RSA algorithm. Perform RSA encryption and decryption with p = 17, q = 11, M = 88. (10)
- VIII. (a) Explain SSL architecture. (5)
- (b) Explain the authentication service provided by PGP. (5)
- OR
- IX. Write short notes on: (10)
 - (i) Intruder (ii) Firewall (iii) Virus (iv) Trojan Horse (v) Worms.